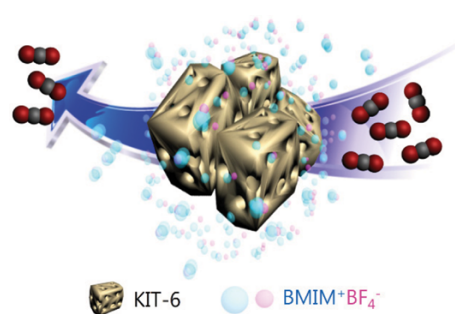


Enhanced Olefin and CO₂ Permeance Through Mesopore-Confined Ionic Liquid Membrane

Il Seok Chae¹Gil Hwan Hong²Donghoon Song¹Yong Soo Kang^{*,1}Sang Wook Kang^{*,2,3}¹Department of Energy Engineering, Hanyang University, Seoul 04763, Korea²Department of Chemistry, Sangmyung University, Seoul 03016, Korea³Department of Chemistry and Energy Engineering, Sangmyung University, Seoul 03016, Korea

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Abstract: Nanocomposites of ionic liquid, 1-butyl-3-methylimidazolium tetrafluoroborate (BmimBF₄), in three-dimensional mesoporous silica (KIT-6) were fabricated and utilized as mesopore-confined ionic liquid membranes for Olefin and CO₂ Separation. Compared to neat BmimBF₄, the fabricated membrane showed 10-times-enhanced gas permeation property for olefin separation, and 5-times-enhanced gas permeation property for CO₂ separation with gas-selectivity remaining unchanged. Moreover, the enhanced diffusion-limited current density of the electrochemical cell involving I⁻/I₃⁻ redox couple in BmimBF₄/KIT-6 nanocomposite was observed. This implies that mesoporous KIT-6 can be used as an effective additive in nanocomposite membranes for enhanced olefin and CO₂ permeance.



Keywords: mesoporous materials, ionic liquids, CO₂, olefin separation, and iodine diffusion.

1. Introduction

Highly ordered mesoporous materials have attracted great interest because of their unique nanostructure with high specific surface area and pore volume.¹⁻³ In particular, three-dimensional (3D) interconnected mesopores have continued to receive tremendous attention due to their nature and access of the pore system. Such mesoporous materials are considered to be highly advantageous for mass diffusion and transportation compared to other mesoporous structures with one- or two-dimensional channels.⁴ However, recent several experimental studies of certain ionic liquids (ILs) under confinement in mesopores have demonstrated a decrease in the diffusivity due to the pore wall-IL interaction.⁵⁻⁸ Conversely, Shi *et al.* predicted an increase of self-diffusion coefficients in ILs confined in carbon nanotubes through molecular dynamics and Monte-Carlo simulations.⁹ Also, Iacob *et al.* presented the experimental evidence of 2 orders-enhancement of molecular mobility and ionic conductivity of a confined 1-butyl-3-methylimidazolium tetrafluoroborate (BmimBF₄) under a mean 7.5 nm pore diameter.¹⁰ Herein, we present the first results of the application of 3D interconnected mesoporous sil-

ica (KIT-6) filled with BmimBF₄ for fabricating highly permeable gas-separation membranes.

For developing high-performance membrane-based gas-separation systems, we recently proposed a carrier-mediated mass transport, *i.e.*, facilitated transport for the separation of olefin/paraffin or CO₂/N₂ mixture.¹¹⁻¹³ In such membranes comprising specific compounds as the gas carriers in polymer matrix or ILs, the permeation performance of the target molecules increases through a carrier-mediated mass transport; consequently, a high selectivity can be obtained. For instance, 1-butyl-3-methylimidazolium tetrafluoroborate (BmimBF₄)/Ag nanocomposite membrane is reported to demonstrate a propylene/propane selectivity of 17 and a total mixed-gas permeance of 2.7 GPU (GPU=gas permeation unit, 1 GPU=1×10⁻⁶ cm³ (STP)/(cm² s cm Hg); STP=standard temperature and pressure).¹¹ We also reported that the olefin-separation performance by neat IL media was mainly due to specific and reversible interactions of ILs such as BmimBF₄ and 1-methyl-3-octylimidazolium nitrate (MoimNO₃) with propylene.¹² Furthermore, imidazolium-based room-temperature ILs have been widely investigated in the separation of not only olefin/paraffin mixture gases but also acidic gases such as SO₂, H₂S, and CO₂ from CH₄ and N₂.¹⁴⁻²³ However, such IL system showed relatively low permeance, which is usually even more challenging than selectivity for industrial applications.

Taking the advantage of both ILs and KIT-6 toward mass-transport phenomena, in this study, we expected that a faster penetrant diffusion should occur in a BmimBF₄/KIT-6 nanocomposite through the confinement effects and a systematic manipulation of the diffusion path. Unusually, *ca.* 10-times-enhanced gas permeation property was demonstrated, with the selectivity remaining unchanged. The enhanced diffusion property of the solute in the media (ILs) was also supported by the diffu-

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***Corresponding Authors:** Yong Soo Kang (kangys@hanyang.ac.kr), Sang Wook Kang (swkang@smu.ac.kr)

sion-limited current density using an electrochemical cell involving I^-/I_3^- redox couple. Thus, the application of BmimBF₄/KIT-6 nanocomposite for the accelerated facilitated transport in olefin/paraffin-, CO₂/CH₄-, and CO₂/N₂-separation membranes was investigated.

2. Experimental

2.1. Preparation of KIT-6

The preparation of mesoporous silica KIT-6 was based on a report of Ryoo's group.²⁴ Pluronic P123 (EO₂₀PO₇₀EO₂₀, MW=5800, as structure-directing agent), tetraethoxysilane (TEOS, as a silica source), HCl, and all the chemicals for preparing KIT-6 were used as received from Aldrich Chemical Co.

2.2. Nanocomposites BmimBF₄/KIT-6 membrane

KIT-6 was added to 1-butyl-3-methylimidazolium tetrafluoroborate (BmimBF₄, obtained from C-TRI, Korea). After stirring the solution of IL and KIT-6 composites for 24 h, the BmimBF₄/KIT-6 nanocomposite coated onto a polysulfone microporous membrane support (Woongjin Chemical Industries Inc., Seoul, Korea) using an RK Control Coater (Model 101, Control Coater RK Print-Coat Instruments Ltd., UK). The selective layer was estimated to be *ca.* 20 μ m using the energy dispersive X-ray (EDX) attached to scanning electron microscopy (SEM) in Supplementary material. The average pore size of the surface of the macroporous membrane support was 0.1 μ m, whereas the thickness of the asymmetric structure was $\sim$ 40 μ m. Ionic liquids, and fumed silica powder (7 nm) was from Aldrich Chemical Co.

2.3. Membrane performances

The BmimBF₄/KIT-6 membrane was directly equipped with a permeation cell. The flow rates of the mixed and sweep gases (helium) were controlled using mass-flow controllers. The all-gas flow rates represented by GPU were determined using a mass flow meter in its steady state. The gas flow rates and gas permeances were measured using a mass flow meter at an upstream pressure of 40 psig and at atmospheric pressure downstream. The 50:50 propylene/propane mixed gas-separation properties of the BmimBF₄/KIT-6 membranes with a membrane area of 2.25 cm² were evaluated using a gas chromatograph (Hewlett-Packard G1530A, MA) equipped with a thermal conductivity detector (TCD) and a unibead 2S 60/80 packed column.

The gas permeances of these membranes for CO₂, N₂, and CH₄ were measured using a bubble meter. The single-gas permeance (*Q*) was measured for CH₄, N₂ and CO₂ at a pressure difference of 2 atm, representing the pressure-normalized mass flux is mathematically expressed as

$$J = Q \times \Delta p$$

where *J* is the mass flux, and Δp is the pressure difference across the membrane. The permeance was determined from the volume of permeate gas. The selectivity of the gas permeance is defined as the ratio of the CO₂/N₂ and CO₂/CH₄ permeance.

2.4. Preparation of the electrochemical cell

For investigation of diffusion-limited current density of BmimBF₄/KIT-6 medium, an electrochemical cell prepared according to Houch's report.²⁵ Symmetrical dummy cells were assembled using two identical Pt-electrodes with the same composition used to fabricate the DSCs. Pt electrode was prepared by spreading a drop of H₂PtCl₆ solution (2 mg Pt in 1 mL ethanol) onto an FTO glass substrate followed by heating at 400 °C for 15 min. The BmimBF₄/KIT-6 electrolyte consisted of 0.6 M 1-butyl-3-methylimidazolium iodide (BMII, Solaronix), 0.05 M iodine (I₂) (Sigma-Aldrich, ACS reagent, 99%).

2.5. Characterizations

Transmission electron microscopy (TEM) images were obtained by JEOL JEM-3000, operating at 300 kV. Samples for TEM imaging were prepared from a copper grid where the solution was dropped. The FT-Raman spectra were collected for the CuNPs/ILs/GO films at room temperature using JASCO NRS-3100 at a resolution of 1.0 cm⁻¹. Cyclic voltammetry (CV; Autolab potentiostat/galvanostat) was performed in symmetrical dummy cells.

Nitrogen adsorption-desorption measurements of porous silicas were performed on an Autosorb 1 instrument (Quantachrome Instruments) at 77 K. Samples were preheated at 120 °C for 3 h under 1×10^{-2} Torr prior to the measurements. The pore size distributions and specific surface areas were evaluated by Barrett-Joyner-Halenda (BJH) and Brunauer-Emmett-Teller (BET) method, respectively. Also, density functional theory (DFT) method using non-local density functional theory (NLDFT) adsorption branch model were used for the textural parameters of KIT-6.

3. Results and discussion

The highly ordered mesostructures of KIT-6 is confirmed by TEM image in Figure 1(a). N₂ adsorption/desorption isotherm is of type IV, which is typical for mesoporous materials in Fig-

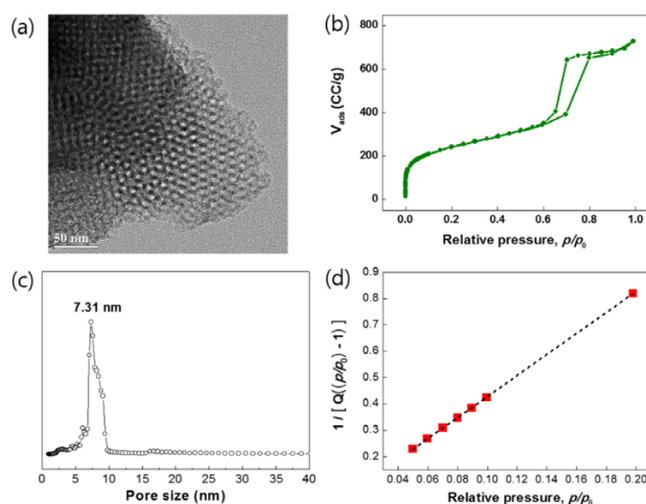


Figure 1. (a) The TEM image of KIT-6. (b) N₂ adsorption-desorption isotherms and corresponding (c) DFT pore size distribution curve, and (d) BET-plot for KIT-6.

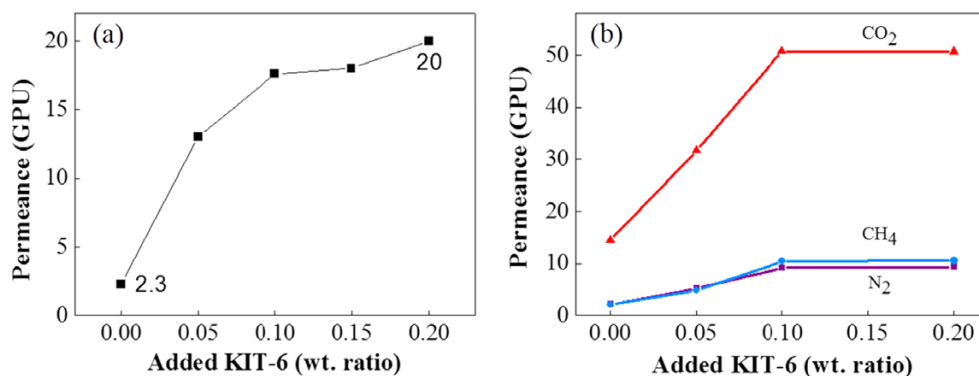


Figure 2. Permeance of BmimBF₄/KIT-6 membranes for separation of (a) propylene/propane (50:50 v/v) mixed gas with the mixed-gas selectivity remained constant at 3.2, and (b) CO₂ from CH₄ and N₂.

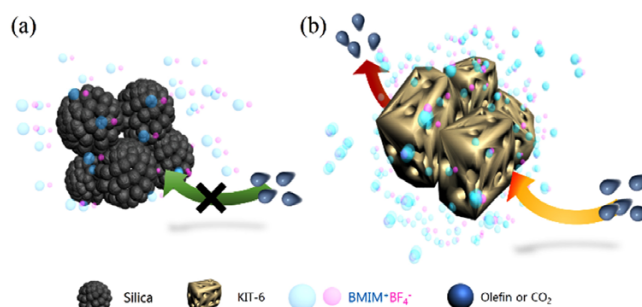
ure 2(b). The pore diameter calculated by the BJH method is 6.62 nm; On the other hand, the DFT model assuming a cylindrical pore geometry gives diameters of 7.31 nm for the main pores and specific surface area of 754 m²/g. Terasaki *et al.* reported that the DFT model was in good agreement with the 3D reconstructed structure, which was consisted by two parts, *i.e.*, main pore and complementary pores.²⁶ The BET surface area is 866 m²/g, which was calculated by the multipoint BET model from the adsorption data, as shown Figure 1(d). As expected, the high structural order of the large-pore cubic Ia3d KIT-6 materials with a narrow mesopore (5~10 nm) distribution was prepared, which are in good agreement with those reported previously for similar samples.²⁴

The propylene/propane (50:50 v/v) mixture-separation performances of BmimBF₄/KIT-6 membranes with various composition of KIT-6 were evaluated. Figure 2(a) shows that the neat BmimBF₄ membrane (*i.e.*, BmimBF₄ membrane without KIT-6) exhibited a relatively low permeance performance: The gas permeance was 2.3. Surprisingly, the introduction of 0.05 weight ratio of KIT-6 to the BmimBF₄ membrane increased the permeance from 2.3 to 13.4, whereas the mixed-gas selectivity remained constant at 3.2 in our previous report. Ultimately, the maximum permeance 20 was obtained in BmimBF₄/KIT-6 (1:0.2 w/w). In parallel, for CO₂ separation, BmimBF₄/KIT-6 (1:0.1 w/w) increased the permeance from 17 to 51; the other gases such as CH₄ and N₂ also increased the permeance from 2.14 to 9.33 and 10.58, respectively. Ideal separation factor and CO₂ permeances are summarized in Table 1.

There have been several challenge to increase the diffusivity of ionic liquid for CO₂ separation. At the similar condition, *i.e.*, liquid membranes using BmimBF₄ on support, for example, Zhao

et al. reported that a small addition of water in BmimBF₄ decreased viscosity of IL, which led to increase CO₂ permeance from 11.5 to 13.8 GPU with CO₂/N₂ selectivity of 60.²⁷ Here, after the addition of KIT-6 into BmimBF₄, notwithstanding the change from liquid-state to quasi-solid-state, the increased diffusivity of the media regarding gas penetration were observed.

Interestingly, the highly enhanced permeance of BmimBF₄/KIT-6 nanocomposite was observed to be better than that of neat BmimBF₄ membrane for the same thickness of selective layer and almost same selectivity performance. We also examined the effect of a mesoporous network of fumed silica in BmimBF₄ for investigating the gas permeance behavior. However, BmimBF₄/



Scheme 1. Fumed silica (a) and KIT-6 (b) in BmimBF₄ for olefin or CO₂ separation.

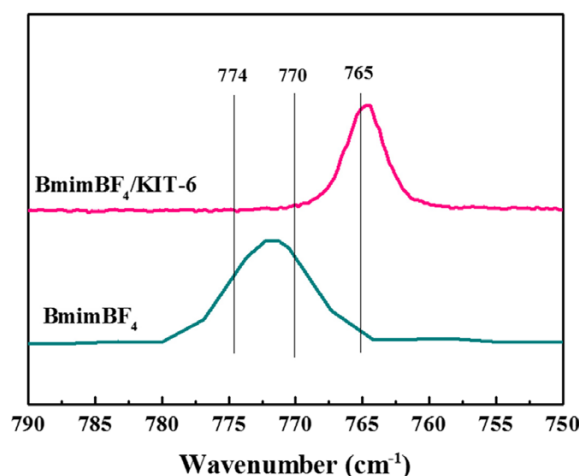


Figure 3. FT-Raman spectra: BmimBF₄/KIT-6 gelled powder composite and neat BmimBF₄ in the BF₄⁻ stretching region.

Table 1. Selectivity and CO₂ permeance of neat BmimBF₄ and various BmimBF₄/KIT-6 composite membranes under 2 atm pressure difference at room-temperature

BmimBF ₄ /KIT-6 composition (w/w)	Selectivity (CO ₂ /N ₂)	Selectivity (CO ₂ /CH ₄)	CO ₂ permeance (GPU)
No KIT-6	5.0	4.8	17.0
1:0.05	6.0	6.6	31.7
1:0.10	5.1	4.2	51.7
1:0.20	5.4	4.8	51.6

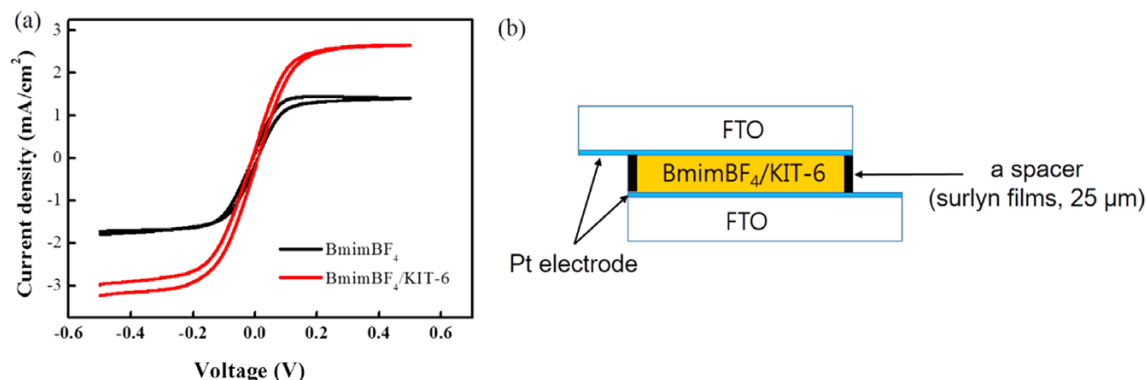


Figure 4. (a) Steady-state current-voltage curve of MPIO-based liquid and BmimBF₄/KIT-6 (1:0.05 w/w) gel electrolytes; scan rate: 5 mV/s, and (b) The electrochemical cell used for the measurements.

silica (1:0.05 w/w) showed a slow gas transport that could not be measured. Similarly, Le Bideau *et al.* reported a decrease in the diffusivity of ILs in monolithic silica matrices.²⁸ Thus, the mesoporous network of fumed silica *without mesopores* may act as a barrier of gas penetration, as shown in Scheme 1.

The interactions between BmimBF₄ and KIT-6 in 1/0.1 BmimBF₄/KIT-6 gelled powder composites were investigated by Fourier transform (FT)-Raman spectroscopy. The Raman spectra in the regions of BF₄⁻ stretching bands are shown in Figure 3. The peaks for free ions, ion pairs, and higher-order ion aggregates of BF₄⁻ appeared at 765, 770, and 774 cm⁻¹,¹¹ respectively. Only the peak intensity at 765 cm⁻¹, corresponding to free BF₄⁻ anions, was observed in BmimBF₄/KIT-6 (1:0.1 w/w). In contrast, in the absence of KIT-6, neat BmimBF₄ showed the almost aggregated and ion-paired states of anions. This increase can be explained by the fact that the introduction of inorganic nanoparticles greatly increases the concentration of free anions. Therefore, the use of KIT-6 as the gelator in BmimBF₄ was successfully demonstrated by the favored interactions between the KIT-6 and BF₄⁻.

Finally, the diffusion media property of BmimBF₄/KIT-6 was evaluated using an electrochemical cell, which was designed based on the report on DSC application as shown in Figure 4 (b).²⁵ The cell was filled with the electrolyte: 0.6 M of 1-methyl-3-propylimidazolium iodide (MPIO) and 0.05 M of iodine (I₂) in BmimBF₄/KIT-6 (1:0.05 w/w) nanocomposite.

Figure 4 shows the comparative steady-state voltammograms for BmimBF₄/KIT-6 obtained at the scan rate of 5 mV/s. The diffusion constant of I₃⁻ was calculated from the diffusion-limited current density (j_{lim}) by using Eq. (1) (Table 2).

$$D_{I_3^-} = \frac{l}{2nFc_{I_3^-}} j_{lim} \quad (1)$$

where n is the electron number per molecule, F is the Faraday constant, and $c_{I_3^-}$ is the concentration of I₃⁻ species. Because of large excess of I⁻ in the system, only the diffusion of I₃⁻ limits the

Table 2. Diffusion-limited current (j_{lim}) and diffusion constant ($D_{I_3^-}$) of I₃⁻ in BmimBF₄ in the absence of KIT-6.

	j_{lim} (mA/cm ²)	$D_{I_3^-}$ (cm ² /s)
BmimBF ₄	1.42	1.84×10^{-7}
BmimBF ₄ /KIT-6	2.66	3.45×10^{-7}

current. As shown in Table 1, the diffusion constant of I₃⁻ ($D_{I_3^-}$) in BmimBF₄ increased from 1.84 to 3.75 when 0.05 weight ratio of KIT-6 was added, which is 87.5% of the increased value and is consistent with the gas-separation performances of the membranes. Therefore, this result clearly showed that by utilizing 3D interconnected mesopores such as KIT-6, hybridization can be used as one of the effective strategies for enhancing the mass-transport properties.

4. Conclusions

In summary, KIT-6 was used as an additive to maximize the permeance performance of highly selective membranes for gas separation. Such increased diffusion properties are mainly attributed to the nano-confined ionic liquid effects and a systematic manipulation of the diffusion path. We believe that such a quasi-solid-state electrolyte offers specific benefits over the ILs for gas separation and provides the new insights in dynamics of IL under confinement.

Supporting information: Information is available regarding the cross section of the BmimBF₄/KIT-6 (1:0.1 w/w) membrane. The materials are available *via* the Internet at <http://www.springer.com/13233>.

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